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MILITARY SPECIFICATION

MANUALS, TECHNICAL: STORAGE AND MAINTENANCE PROCEDURES (FOR NONNUCLEAR AND CHEMICAL MUNITIONS/BIOLOGICAL RESEARCH AGENTS)

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the requirements/format for preparation of technical manuals for inspection, testing, maintenance, and storage procedures to be accomplished during storage of nonnuclear and chemical munitions/biological research agents.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Specifications

Military

MIL-N-7513	Nomenclature Assignment, Contractors Method for Obtaining
MIL-M-38784	Manuals, Technical, General Style and Format Requirements

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC/MMAPD, Wright-Patterson AFB OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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MIL-P-38790 Printing Production of Technical Manuals, General Requirements for

MIL-M-38807 Manuals, Technical, Illustrated Parts Breakdown, Preparation of

Standards

Military

MIL-STD-129 Marking for Shipment and Storage

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specific .. to the extent specified herein.

Technical Orders

00-5-1 Air Force Technical Order System

11A-1-1 Ammunition, Restricted or Suspended

11A-1-10 General instructions, Ammunition Service ability procedure

11A-1-46 Fire Fighting Guidance, Transportation and Storing, Management Data and Ammunition Complete Round Chart

11A-1-53 General Instructions for Ammunition Color Coding and Identification of Empty and Inert Loaded Ammunition Items and Components, and Assignment of Version Numbers to Training and Dummy Ammunition Items

42A-1-1 Safety, Fire Precaution and Health Promotion Aspects of Painting, Doping and Paint Removal

Air Force Regulations

0-17 Index of AFOSH Standards

127-12 Occupational Safety and Health Program

127-100 Explosives Safety Standards

Air Force Manuals

AFM 67-1 USAF Supply Manual

DoD Directives

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Preparation. The general manner of preparation shall be in accordance with MIL-M-38784 and MIL-P-38790.

3.1.1 Requirements to be included. Manuals shall include all procedures necessary to establish a complete progressive storage and maintenance program. No requirement shall be omitted even if the requirements would not be applicable until expiration of a long interval of storage. Work described shall be limited to that which is accomplished in the designated storage, inspection and maintenance areas. Terms not readily understood shall be defined as they are used. Requirements shall be based on the contractor's recommendations and service experience. User recommendations will be considered in the inclusion of requirements. Operation and maintenance instructions for auxiliary equipment shall not be included; applicable manuals shall be referenced.

3.1.2 Stating requirements. Procedures shall be stated briefly. Each storage, handling and inspection requirement shall specifically state what condition is to be sought.

3.1.3 Identification of items. The official nomenclature in accordance with MIL-M-7513 and Department of Defense Identification Code (DODIC), if assigned, of items shall be listed one time in the manual, in Table 2-1. Further identification shall be in short descriptive form, but definitive enough to ensure clear identification.

3.2 Nonnuclear munitions manual. Nonnuclear munitions manuals shall have the following arrangement in the order indicated. When a chapter is not applicable, it shall be omitted and succeeding chapters renumbered accordingly.

NONNUCLEAR MUNITIONS

Front Matter

Chapter 1	Safety and Accident Prevention
Chapter 2	Description
Chapter 3	Tools, Equipment and List of Consumables
Chapter 4	Handling and Storage
Chapter 5	Inspection and Testing

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Chapter 6 Preparation for Use, Maintenance and Storage

Chapter 7 Illustrated Parts Breakdown

3.2.1 Front matter. The requirements of MIL-M-38784 are applicable

3.2.1.1 Foreword. The foreword shall be in accordance with MIL-M-38784. The definition of "shall", "will", "should" and "may" contained in MIL-M-38784 shall be included. The definition of "WARNINGS", "CAUTIONS" AND "NOTES" contained in MIL-M-38784 shall be included. The following statement shall be included: "Improvement Reports (AFTO Form 22) for this Technical Order shall be forwarded to (enter the responsible agency and address) in accordance with TO 00-5-1".

3.2.1.1.2 Chapter 1, safety and accident prevention. General safety requirements shall be in the form of general text information. Data shall provide safety and accident prevention guidance necessary to safely utilize or handle the munitions items. This information shall be in accordance with the provisions of AFR 127-100, AFR 127-12 and AFOSH standards as listed in AFR 0-17. Safety Requirements shall be included even though the manual covers only nonexplosive items. (see figure 1). The following paragraph(s) shall be included, as applicable:

a. If a transmitting antenna of radar or other such generating devices are in the area, consult AFR 127-100. The correct safety distances for electro-explosive devices will be followed.

b. When spray painting, personnel shall wear National Institute of Occupational Safety and Health (NIOSH) approved organic vapor cartridge type respirators, unless otherwise advised by bioenvironmental engineer.

c. When using approved agents for minor cleaning operations, personnel shall routinely use rubber gloves and chemical goggles and/or face shield. All ventilation shall meet the requirements of AFOSH standard 161-2. Consult local bioenvironmental engineer to determine respiratory protection requirement.

d. The provisions to T.O. 42A-1-1 will be strictly adhered to when performing any operations other than those listed in b and c above. All ventilation shall meet the requirements of AFOSH Standard 161-2.

e. Personnel shall wear eye protection and gloves when installing or removing banding.

f. Personnel shall promptly report safety, fire, and health hazzards, operational injuries, and use required protective clothing or equipment.

3.2.2 Chapter 2, description. This chapter shall have the following arrangement in the order indicated.

General

Item description

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CHAPTER 1
SAFETY AND ACCIDENT PREVENTION

1-1. GENERAL. These safety requirements will be complied with during all operations involving explosives. All personnel engaged in operations involving explosives and/or other hazardous material shall be thoroughly trained in safety. They shall be capable of recognizing hazardous situations and conditions. Thinking safety and working safely must become a habit when working with or near items that can create a hazard due to their explosive, flammable, or toxic fillers.

1-2. REQUIREMENTS. Safety requirements are as follows:

1-2.1. The Explosives Safety Standards AFR 127-100 will be followed. Lack of a safety requirement in this technical manual or the above reference does not mean that no safeguards are needed. Prompt action will be taken to control any hazard. If a dangerous item is encountered all operations in the immediate area will be shut down. Personnel will be evacuated to a safe site. EOD or other authorized personnel will be called for help in eliminating the hazard. Operations will not be resumed until the hazard has been eliminated.

1-2.2. If a transmitting antenna of radar or other such generating devices are in the area, consult AFR 127-100. The correct safety distances for electro-explosive devices will be followed.

1-2.3. When an abnormal condition is noted and procedures are not available in this manual, work will be stopped and qualified guidance obtained.

1-2.4. When spray painting, personnel shall wear National Institute of Occupational Safety and Health (NIOSH) approved organic vapor cartridge type respirators, unless otherwise advised by bioenvironmental engineer.

1-2.5. When using approved agents for minor cleaning operations, personnel shall routinely use rubber gloves and chemical goggles and/or face shield. All ventilation shall meet the requirements of AFOSH standard 161-2. Consult local bioenvironmental engineer to determine respiratory protection requirement.

1-2.6. The provisions to T.O. 42A-1-1 will be strictly adhered to when performing any operations other than those listed in b and c above. All ventilation shall meet the requirements of AFOSH Standard 161-2.

1-2.7. Personnel shall wear eye protection and gloves when installing or removing banding.

1-2.8. Personnel shall promptly report safety, fire, and health hazzards, operational injuries, and use required protective clothing or equipment.

FIGURE 1. Sample Chapter 1.

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TABLE 2-I. Leading Characteristics

NOMEN-CLATURE (1)	FSC/ DODIC (2)	PART NO. AND/OR MODEL NO./ FSCM	DIMENSIONS (INCHES) (3)				WEIGHT (POUNDS)		DIMENSIONS (INCHES) OUTER PACK
			LENGTH	WIDTH	HEIGHT	DIAMETER	GROSS (4)	NEW (ONE ITEM) (5, 6)	

NOTES

1. The nomenclature appearing in the title block of the drawing to which the detail part was manufactured shall be used. Short form or normal reading order may be used throughout the text as long as positive identification is maintained.

2.. DODIC will be used when available.

3. Use dimension columns as appropriate.

4. Weights under one pound given to the ten thousandth of a pound (i.e., .0001, .5000 etc). Weights over one pound to the nearest tenth (i.e., 24.5). Weights will be rounded to the next highest number (i.e., 0.00013 would be shown as 0.0002 and 13.83 would be shown as 13.9).

5. NEW and flat note will be placed in weight column and flag note with definition will be at the bottom of table on each sheet if there are more than one.

6. Net Explosive Weight (NEW) in accordance with TO 11A-1-46.

FIGURE 2. Table of Leading Characteristics.

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Physical description

Functional description

Packaging

Marking

3.2.2.1 General. A brief explanation of the contents and scope of this chapter shall be included. The last sentence shall reference a table of leading characteristics (see figure 2).

3.2.2.2 Item description. Separate paragraphs will be used for each service (end) item, training/dummy item and major components covered in the manual.

3.2.2.2.1 Physical description. A physical description of the general construction and purpose of the item and illustration(s) that will portray the physical characteristics. Features mentioned in the text shall be called out on the illustration(s).

3.2.2.2.2 Functional description. A detailed functional description of the service (end) item and a cut-away view to supplement the description shall be included. Features mentioned in the text shall be called out on illustration(s).

3.2.2.3 Packaging. This paragraph(s) or section shall provide description and illustrations of the original shipping and storage containers for the service (end) item, training/dummy item and major components (as applicable). Special packaging instructions/transportation packaging orders (SPI/TPO) will be used as the source document. Alternate packaging may be included. As a minimum, this data shall contain:

a. Description and illustrations of packaging and overpack requirements.

b. Illustrations of the end item and major components packaging markings.

c. Original SPI/TPO drawing number.

3.2.2.4 Marking. Illustrate the marking requirements for the service (end) item, training/dummy item, major components and containers (as applicable). The physical description illustration(s) shall be used to show markings. Container markings will be shown on the container illustration(s). Color coding requirements will be included.

3.2.2.5 Table of leading characteristics. A table of leading characteristics will be prepared in the format shown in Figure 2 . All items covered by the manual will be included in the table.

3.2.3 Chapter 3, common safety, special tools, test equipment and list of consumables. This chapter shall list and illustrate the common safety, special tools and test equipment required for the work described. Such items shall be selected from the list of tools and test equipment approved by the Government by means of support equipment (SE) listings, provisioning conferences, Engineering Change Proposals (ECPs), procurement documents. Standard types of tools, such as screwdrivers and pliers shall not be listed. Standard types of test equipment, such as voltmeters and tube testers shall be listed. When a contractor cannot obtain an approved list, a manual shall show the contractor's recommended special tools and test equipment. However, these recommendations shall be changed promptly, if necessary, to conform to the official Government list as soon as such information is provided to the contractor.

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3.2.3.1 Common safety and special tools list. The list of common safety and special tools shall be arranged in the following format. Tools not illustrated in use in other chapters of the manual shall be illustrated following the list in this chapter. A warning shall be included prohibiting use of unauthorized substitutes.

SPECIAL TOOLS LIST

Part (Tool) Number and National Stock Number for Special Tools only.	Figure & Index No.	Nomenclature	Application

a. Part (tool) number. Tools shall be listed in alpha-numeric sequence by part (tool) number and manufacturer's five digit code or manufacturer's name and address. The National Stock Number (NSN) for special tools shall be listed below the part number and FSCM.

b. Figure and index number. The figure and index numbers that appear in other chapters, or this chapter, of the manual, shall be shown.

c. Nomenclature. Nomenclature shall be in accordance with the requirements of MIL-M-38784.

d. Application. The application of each tool shall be stated by inserting one or more of the following application codes.

APPLICATION CODES

A - Assembly	H - Handling	I - Inspection	M - Maintenance
S - Storage	T - Test		

3.2.3.2 Test Equipment List. The list of test equipment shall be arranged in the following format. Test equipment not illustrated in use in other chapters of the manual shall be illustrated following the test equpiment list in this chapter (standard test equipment excepted).

TEST EQUIPMENT LIST

Type Designation	Alternate Type Designation	Figure & Index No.	Nomenclature	Application

a. Type designation. Test equipment shall be listed in alpha-numeric sequence by type designation, using the AN type designation, if assigned and the manufacturer's five digit code or manufacturer's name and address. Special/safety related test equipment, shall have the NSN listed below the type designation.

b. Alternate type designation. If an alternate item of test equipment can be used, its AN type designation, if assigned, or the commercial or manufacturer's designation shall be listed.

c. Figure and index number. The figure and index numbers that appear in other chapters, or this chapter, of the manual, shall be shown.

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d. Nomenclature. Nomenclature shall be in accordance with the requirements of MIL-M-38784.

e. Application. The application of each item of test equipment shall be stated by inserting one or more application codes (see 3.2.3.1).

3.2.3.3 Consumables. Expendable items and support materials (consumables) shall be listed. Sealants, lubricants, gaskets, seals, cleaning solvents, paint, etc., are considered expendable items. The list will be in tabular form, in alphabetical order by nomenclature, as follows:

Nomenclature	Specification(s)	Part number and FSCM	Application (see 3.2.3.1 for (Application code)

3.2.4 Chapter 4, handling and storage. This chapter shall have the following arrangement in the order indicated. (see figure 3.)

General

Handling

Storage Data

Storage requirements

3.2.4.1 General. Principal hazards involved in storage and handling of items and explosive components listed in this manual will be included. Terms such as shelf life, service life, magazine, and any other items used shall be defined and their use explained. An identification subparagraph will be included to read: "The use of standard nomenclature and lot number/serial number is mandatory for all purposes of storage records, reports and communications concerning munitions. Legible identification markings will be maintained on munitions in storage."

3.2.4.2 Service Life. This paragraph shall state when the service life starts and stops, i.e., upon opening container, upon installation, etc., service life for items packaged in hermetically sealed containers service starts on the date of opening and continues until items are expended or expired.

3.2.4.3 Handling. Instructions shall include handling and specialized handling procedures to include acceptable drop criteria. This data will be taken from the manufacturing specifications and will include drop heights for packaged and non-packaged items in the format of a caution. A warning pertaining to careful handling of explosive material will be included. The following statement will be included: "Damaged containers will be repaired as soon as possible and markings restored to the repaired area in accordance with chapter 2. Chemical protection equipment required when handling items covered by the T.O. shall be in accordance with AFR 127-100."

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CHAPTER 4

HANDLING AND STORAGE

4-1. GENERAL. Compliance with AFR 127-100 and the instructions in this manual will ensure safe handling, storage, and serviceability of _____. Waivers and deviations will be in accordance with AFR 127-100. Stored _____. should be protected from adverse climactic conditions. The main hazards linked with the storage and handling of items listed in this TO are:

a. Blast.

b. Fragments

c. Fire.

4-2. SPECIAL TERMS. The following Terms, as defined, apply to _____.

NOTE

Shelf and service lives are not cumulative. Any combination of shelf and service life accrued by an item cannot exceed the shelf life.

4-2.1. Shelf life: The length of time an item can remain in storage. The expiration date for shelf life on items with the month and year listed is the last day of the month.

4-2.2. Service life: The length time an item can remain in operating configuration or in actual usage.

NOTE: For those items packed in hermetically sealed containers service life starts on date of opening and continues until item(s) are expended or expired.

NOTE: For items covered by this T.O., service life starts upon initial installation and runs to expiration even if items is subsequently removed and returned to storage.

4-3. IDENTIFICATION. The use of standard nomenclature and lot number/ serial number is mandatory for all storage records and communications. Legible identification markings will be kept on munitions in storage.

4-3. HANDLING.

WARNING

_____ contain explosives and will not be handled roughly or dropped as personal injury could result. (Other can information can be added).

CAUTION

Packaged _____ dropped more than ___feet or an unpackaged____dropped more than __ feet shall be considered unserviceable and shall be reported in accordance with AFM 67-1, Vol I, Part 1, Chap 20. (Use only if applicable). Damaged containers will be repaired as soon as possible and markings restored to the repaired area.

4-5. STORAGE DATA. For storage data, see Table 4-1, TO 11A-1-46 and DOD ammunition catalog.

4-6. STORAGE REQUIREMENTS. The following requirements are for storage of _____.

FIGURE 3. Sample Chapter 4.

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4.6-1. General Requirements.
a.___shall be stored and stacked by type, lot number, condition code quantity distance requirements, and compatibility group.

b. Different lots shall be stacked separately, and the separation between lots clearly defined.

c. Incomplete packages shall be securely closed and properly marked in accordance with TO 11A-1-10 to identify contents and quantity. Only one lite box will be permitted for each lot and condition code per storage location and shall be stored on the top and front of the stack.

d. Runners, skids or intergral dunnage affixed to containers shall be considered part of required dunnage.

4-6.2. Inside requirements.

Preferred storage is in structures suitable for storing____. The following requirements apply to inside storage:

NOTE

a. Dunnage shall provide a minimum clearance of two inches from the floor.

b. Heat shall not be provided by stoves or open fires.

4-6.3. Outside requirements.____ shall only be stored outside when inside storage is not available. When outside storage is necessary shelter shall be provided as follows:

a. When possible, build a fram- material and allow an 18-inch clearance overhead and on all sides. Cover the framework with fire resistant, water-proof material.

b. Dunnage should provide a minumum clearance at the bottom of the stacks follows:

(1) On hardstand, four inches.
(2) Unimproved surface, six inches.

4-7. PRIORITY OF ISSUE.
To determine the priority of issue of items in storage the following should be considered:

a. Oldest item (if shelf life is indefinite, items with least favorable storage have precedence).

b. Items which have had the least favorable storage.

c. Lite box/container.

d. All others (i.e., Cond code B&C items, service life started, etc.).

FIGURE 3. Sample Chapter 4. (Continued)

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NOTE 1	NOTE 2
(Munition items with an established shelf life). When the lot number or date of manufacture is not known, an item shall not be issued. If the lot number is known the date of manufacture may be obtained from OO-ALC/MMWRA.	(Munition items with an indefinite or N/A shelf life). When the lot number is not known an item will not be issued. Missing date of manufacture will not be cause for classifying an item unserviceable.

FIGURE 3. Sample Chapter 4. (Continued)

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3.2.4.4 Storage data. Storage data will include all requirements that will be used when storing the item. Data will be presented in table format as follows and shall be Table 3-1:

	TEMPERATURE LIMITATIONS (OF)		LIFE	
ITEM	MIN	MAX	SHELF	SERVICE
All munitions/components, (nomenclature, type, model, and part number)	(if applicable)		Years, months or days as Applicable	

3.2.4.5 Storage requirements. Data will cover general storage requirements, priority of issue, serviceability statement, and requirements pertaining to both inside and outside storage. Stacking limitations will be included, if applicable.

3.2.4.5.1 Priority of issue. The paragraph will contain notes pertaining to either or both established or indefinite shelf life items. The note for established shelf life items will read: "When the lot number or date of manufacture cannot be identified, an item shall not be issued. If the lot number or date of manufacture is known, the other may be obtained from OO-ALC/MMWR." The note for indefinite shelf-life items will read: "When the lot number cannot be identified, an item will not be issued. Missing date of manufacture is not significant and will not be cause for classifying the item unserviceable for issue. If only the lot number or date of manufacture is known and the unknown information is needed, it may be obtained from OO-ALC/MMWR."

3.2.4.6 Dunnage requirements. Dunnage requirements will be as shown in figure 3.

3.2.5 Chapter 5, inspection and testing. This chapter shall have the following arrangement in the order indicated. When a paragraph is not applicable, it shall be omitted and succeeding paragraphs numbered accordingly. (see figure 4.)

a. Inspection procedures

(1) Receiving inspection

(2) Shipping inspection

(3) Periodic inspection

(4) Preissue inspection

(5) Combat/simulated combat preissue inspection (if applicable)

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CHAPTER 5

INSPECTION AND TESTING

5-1. GENERAL. Inspection (and testing, if applicable) will be performed to ensur the serviceability of _____ in storage or issued to users. Inspections will be limited to visual inspections. Lot numbers will be checked against TO 11A-1-1 for current status of the munitions.

5-2. DEFECTS. Defects found during inspections will be handled as follows:

5-2.1. Correctable defects._____ with correctable defects will be repaired immediately or temporarily rejected. They will be segrated, marked defective, and scheduled for maintenance. Corrective action is listed in Table 5-1 of Chapter 5 for all correctable defects.

5-2.2 Noncorrectable defects. ____ with noncorrectable defects will be rejected and segregated. They will be marked defective and reported for disposition in accordance with AFM 67-1.

The statement, that hermetically sealed containers will not be opened during inspections, will be included inspection criteria as applicable. When hermetically sealed containers are not opened, only the containers will be inspected in accordance with the classification of defects table. If the container is defective to the extend that the item may not be adequately protected, the containers will be opened and the item inspected.

5-3. INSPECTION PROCEDURES. Inspections will be performed as follows:

5-3.1 Receiving inspection (RI). A receiving inspection shall be performed upon receipt of shipment. If the inspection reveals defects, proper action will be taken. Complete discrepancy reports in accordance with TO 11A-1-10.

5-3.2 Shipping inspection (SI). A shipping inspection shall be performed whenever a shipment is to be made.

5-3.3 Periodic inspection (PI). A periodic inspection will be performed on a one, three or five year basis according to the national stock class table in TO 11A-1-10. Determine sample table for _____(specify), in TO 11A-1-10. Visually inspect for defects listed in table 5-1.

5-3.4 Preissue inspection (PII). Prior to issue, items will be inspected under the criteria of the PII instructions contained in TO 11A-1-10.

5-3.5 Stroage monitoring inspection (SMI). A storage monitoring inspection will be performed to measure/monitor a specific condition of characteristics.

5-3.6 Returned munitions inspection (RMI). A returned munitions inspection will be performed on items returned from custody/consumption acounts prior to return to other than ready storage. Inspection will be conducted as outlined in TO 11A-1-10.

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5-3.7 Special inspection (SPI). Special inspections may be performed by direction of local authority, major commands, or OO-ALC. Special inspection shall be performed in accordance with specific instructions from the directing authority.

5-4. INSPECTION EVALUATION. Determine the number of allowable defectives using Sampling Table for ____ (specify) in TO 11A-1-10. If a critical defect is found, or if defectives exceed the allowable limits, inspect the lot 100 percent. Defectives will be culled out handled in accordance with paragraph 5-2. The rest of the lot will be returned to serviceable status.

5-5. TEST PROCEDURES. No destructive tests are authorized for __. Special destructive tests may be directed by OO-ALC in support of malfunction investigations or others conducted in accordance with instructions from the directing authority.

FIGURE 4. Sample Chapter 5. (Continued)

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(6) Storage monitoring inspection (if applicable)

(7) Returned munitions inspection

(8) Special inspections

(9) Inspection evaluation

b. Test procedures

3.2.5.1 Inspection procedures. Inspection requirements shall cover packaging, and item, and all components listed which are packaged and stored separately. Inspection requirements for training, dummy/inert and nonexplosive item will be listed as applicable. General guidelines for inspections are contained in TO 11A-1-10 and shall be used as reference in preparing specific requirements. Test requirements will be included as applicable. (see figure 4).

3.2.5.1.1 Receiving inspection (RI). Inspection criteria will be established. General inspection guidelines are contained in TO 11A-1-10 and shall be used as a guide. Checkpoints required to perform that inspection will be presented in the classification of defects table.

3.2.5.1.2 Shipping inspection (SI). Inspection criteria will be established. General inspection guidelines are contained in TO 11A-1-10 and shall be used as a guide. Checkpoints required to perform that inspection will be presented in the classification of defects table.

3.2.5.1.3 Periodic inspection (PI). Periodic inspection intervals will be established by the acquiring activity. Inspections will be performed on a sample size basis using table supplied by the acquiring activity with the number of samples selected in accordance with the sampling table(s). Inspection criteria will be presented in the classification of visual defects table which will include all checkpoints necessary to ensure serviceability of the item(s). Specify items which are exempt from this inspection if applicable. If inspection of empty/inert training items is different than that required for live items, indicate inspection requirements. Guidelines in TO 11A-1-10 shall be used.

3.2.5.1.4 Preissue inspection (PII). Inspections shall provide data required to ensure serviceability of a component prior to issue Items will be inspected by sample size with reference made to the appropriate sample table in TO 11A-1-10. Specific inspection procedures will be as directed in TO 11A-1-10.

3.2.5.1.5 Combat/simulated combat preissue inspection (CPII). If applicable, this inspection will be included to be used during a combat/simulated combat situation. Inspection will include the minimum essential inspection item checkpoints to ensure mission accomplishment. The data will be presented in the table format shown in 3.2.5.2.1.

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3.2.5.1.6 Storage monitoring inspection (SMI). Storage monitoring inspections are performed to measure/monitor a specific condition or characteristic; e.g., humidity indicator check. The inspection, if applicable, will include all instructions necessary to accomplish this inspection. Guidelines contained in TO 11A-1-10 shall be used.

3.2.5.1.7 Returned munitions inspection (RMI). This inspection shall be performed on munitions returned from custody/consumption accounts prior to return to other than ready storage. Inspection criteria will be specified and reference to the guideline in TO 11A-1-10 shall be made.

3.2.5.1.8 Special inspections (SPI). (see figure 4).

3.2.5.2 Tables.

3.2.5.2.1 Classification of Visual Defects. These tables shall identify inspection requirements by component, checkpoint, defect classification (critical, major, or minor) and corrective action. For munition item TOs, other than missile and life support items, Work Unit Codes, from the applicable -06 Work Unit Code Manual, will be added to the Component Column and How Malfunctioned Codes will be added to the Defect Classification Column. This table shall be table 5-1 and shall be in the format below:

Component (Work Unit Code)	Checkpoint	Defect Class (How mal Code)	Corrective Action

3.2.5.3 Inspection evaluation. Inspection evaluation criteria will be presented. Criteria in TO 11A-1-10 may be used as a guideline or reference to TO 11A-1-10 may be made. If tests are included in the manual, this paragraph will follow test requirements and the title of the paragraph will be "Inspection and Test Evaluation."

3.2.5.4 Test procedure. Data shall include a step-by-step procedure using specified equipment for testing munition items (electrical, mechanical, functional, etc). Electrical continuity/resistance checks will include meter preparation procedures (if applicable), protective shielding, electrical grounding, immobilization tie-down requirements, etc.

3.2.6 Chapter 6, preparation for use, maintenance and storage. This chapter shall contain procedures for unpacking, preparation for use, authorized maintenance, preparation for storage, repacking and sealing. A list of materials to be used shall be included. This chapter shall be arranged in the following order:

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a. General

b. Unpackaging

c. Preparation for use

d. Maintenance instructions

e. Preparation for storage

f. Repackaging and sealing

g. Remarking

3.2.6.1 Unpackaging. These instructions shall include procedures for uncrating, opening special containers, removal of components(s), and handling of packing materials. Procedures for storing empty reusable containers shall be included. Procedures for unpackaging items packaged in simple hermetically-sealed containers, or simple one-time use containers, need not be included. Include instructions for marking components/containers when shelf/service life is affected by unpackaging. The following paragraph shall be included: "The safety requirements of AFR 127-100 shall be followed when unpackaging. Before opening a container, ensure that all markings are legible. Record markings that will be destroyed by opening the container. Retain serviceable packaging for possible reuse."

3.2.6.2 Preparation for use. These instructions shall include all procedures for preparing an item for use, such as inspection of components, assembly (buildup), installation of components, alignment, test and complete round inspection, as applicable. When preparation for use/assembly results in change of identification/DODAC code, procedures will be included to ensure all applicable functions are notified.

3.2.6.3 Maintenance instructions. These instructions shall include all procedures authorized for accomplishment at the level of maintenance determined by the acquiring activity and will include those expanded step-by-step procedures for corrective actions in chapter 5, Classification of Defect Tables.

3.2.6.4 Preparation for storage. These instructions shall include all procedures for preparing the item for storage and shall include disassembly, removal of components, safing, etc. If disassembly is reverse of assembly, a statement to this effect shall be included.

3.2.6.5 Repackaging and sealing. When repackaging and sealing is the reverse of unpackaging, a statement to this effect shall be included; otherwise step-by-step procedures shall be included. When heat or wax sealing equipment is required, step-by-step procedures for conducting the operation shall be included. New packaging, when necessary, will be constructed/acquired in accordance with the latest transportation packaging order or special packaging instruction. The Packaging Support Branch, OO-ALC/DSTDM, will be contacted for packaging instructions when packaging data is not defined in a TPO/SPI drawing. As a minimum repackaging and sealing shall contain the following:

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a. Repackaging. Repackage in original containers if possible. If not available, a new container will be constructed/acquired in accordance with latest TPO/SPI drawing number (insert TPO/drawing number). If the container or pack prescribed by the TPO/SPI drawing is unavailable or damaged and cannot be built by the base or obtained through Base Supply, OO-ALC/DSTD, will be contracted. DSTD will either furnish the TPO/SPI pack or grant a waiver and specify an alternate pack for safe shipment of the item.

b. Sealing. Containers will be sealed in accordance with the applicable TPO/SPI or step-by step procedures shall be included, as applicable.

c. Remarking. Remarking shall be completed in accordance with the applicable TPO/SPI or MIL-STD-129.

3.2.6.6 Remarking. These instructions shall include procedures for remarking obliterated or incorrect markings on items or containers. Identification of type paint, stencil, pencil, etc. (Government specification, standard, drawing number, etc) required for remarking will be listed in Table 6-1.

3.2.7 Chapter 7, illustrated parts breakdown. An illustrated parts breakdown shall be included when specified by the acquiring activity. This chapter will be prepared in accordance with MIL-M-38807.

3.3 Chemical munitions/biological research agents.

3.3.1 Front matter and chapter 1 thru 5. These portions of the manual shall be prepared in accordance with the requirements of 3.2.1 through 3.2.5.4.

3.3.2 Chapter 6, preparation for use. This chapter shall include step-by-step procedures for unpacking the munition (if applicable) and preparation of the munition for use. Special safety/precaution to be observed shall be described.

3.3.3 Chapter 7, filling. This chapter shall include step-by-step procedures for filling, component selection and installation, preoperational check of filling equipment, and filling method, if applicable.

3.3.4 Chapter 8, maintenance. This chapter shall include step-by-step procedures for all maintenance authorized for accomplishment within the scope of the manual.

3.3.5 Chapter 9, decontamination. This chapter shall include step-by-step procedures for decontamination of the chemical munitions/biological research agents (CM/BRA). When this information is included in general, or other technical manuals, such manuals shall be referenced. If the agent or decontamination solution is classified, the procedures shall be included in a separate supplement. Procedures shall be arranged in the following order, as applicable.

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a. General

b. Symptoms

c. First Aid

d. Decontamination of personnel

e. Decontamination of clothing

f. Decontamination of equipment

g. Preparation, sampling, and testing of decontamination solution

h. Minor leaks in CM/BRA items

i. Major leaks in CM/BRA items

j. Decontamination of CM/BRA items

k. Chemical or biological research agent spills

3.3.5.1 General. A general description shall be included of the chemical/biological research agent and the decontaminant(s).

3.3.5.2 Symptoms. Symptoms shall be listed which indicate that personnel involved have been exposed to the chemical/biological research agent.

3.3.5.3 First aid. First aid instructions shall be provided for treatment of personnel who have been contaminated by a chemical/biological research agent.

3.3.5.4 Decontamination of personnel. Procedures shall include the steps required for decontamination when wearing impermeable protective clothing and also the steps required for decontamination of personnel not wearing protective clothing.

3.3.5.5 Decontamination of clothing. Procedures for decontaminating clothing which has been exposed to the chemical/biological research agent shall be given.

3.3.5.6 Decontamination of equipment. Instructions shall include all procedures for the decontamination of equipment involved in minor/major leaks and spills of a chemical/biological research agent.

3.3.5.7 Preparation, sampling and testing of decontamination solution. Procedures to prepare, sample and test the decontamination solution shall be provided.

3.3.5.8 Minor leaks in CM/BRA items. The corrective procedures to follow when minor leaks occur in the CM/BRA items shall be given.

3.3.5.9 Major leaks in CM/BRA items. The corrective procedures to follow when major leaks occur in the CM/BRA items shall be given.

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3.3.5.10 Decontamination of CM/BRA items. Procedures to follow in decontaminating the CM/BRA items after completion of a mission, prior to storage or reuse, as applicable, shall be given.

3.3.5.11 Chemical or biological research agent spills. Corrective procedures to follow in decontaminating the area after a chemical or biological research agent spills shall be given.

3.3.6 Chapter 10, preparation for storage. This chapter shall include, as applicable, the procedures for preparing the items for storage in shipping container(s), including repackaging. Procedures for both inert items and special items shall be given. The instructions shall be arranged in the following order:

a. Internal component preparation

b. External component preparation

c. Chemical Agent, Biological research agent and residue removal

d. Seam and area taping

3.3.6.1 Internal component preparation. Instructions shall cover the removal and replacement of any internal components that are to be accomplished prior to storage.

3.3.6.2 External component preparation. Instructions shall be included for removal and replacement of any external components that shall be accomplished prior to storage.

3.3.6.3 Chemical agent, biological research and residue removal. Instructions shall be included for the removal of a chemical agent/biological research agent or the agent residue from the item.

3.3.6.4 Seam and area taping. The seams and areas to be taped prior to storage shall be described. This tape shall be properly identified by part or military specification number. An illustration shall be provided to facilitate location of the taping areas.

3.3.6.5 Chapter 11, illustrated parts breakdown. An illustrated parts breakdown group shall be included when specified by the acquiring activity and will be prepared in accordance with MIL-M-38807.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality. The quality assurance provisions shall be in accordance with MIL-M-38784 and MIL-P-38790.

5. PREPARATION FOR DELIVERY

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5.1 Packaging. Packaging, packing and marking for shipment shall be in accordance with MIL-M-38784 and MIL-P-38790.

6. NOTES

6.1 Intended use. Manuals prepared in accordance with this specification are intended for accomplishing storage and maintenance procedures of nonnuclear munitions and chemical munitions/biological research agents.

6.2 Definitions.

6.2.1 Noncuclear munitions/explosives. Includes all ammunition, munition fillers, demolition material, solid rocket motors, liquid propellants, cartridges, pyrotechnics, mines, bombs, grenades, warheads of all types, explosive elements of ejection/egress and aircraft systems, explosive components of missile systems and space systems, and assembled kits and devices containing explosives material.

6.2.2 Chemical munitions. For the purpose of this specification, a chemical munition is a nonnuclear munition in which the filler is primarily a chemical agent. Chemical agents do not include riot control agents, chemical herbicides, smoke and flame.

6.2.3 Biological research agents. For the purpose of this specification, biological research agents are those small quantities of biological agents (including toxins) necessary for research and development of defense measures and materiel, such as, immunizations, detection and warning devices, and other safety measures.

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirement title	Applicable DID no.	Option
3.2	Technical Orders	DI-M-3407	Storage and Maintenance procedures for nonnuclear munitions
3.3	Technical Orders	DI-M-3407	Storage and Maintenance procedures for chemical munitions/biological research agents.

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(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect tothe previous issue due to the extensiveness of the change.

Custodian:

Air Force - 16

Preparing Activity

Air Force - 16

Review Activities:
Air Force - 01, 10, 70, 99

User Activities:
Air Force - 11, 15, 18, 71, 84, 79, 80, 82,

(Project TMSS-F466)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38781E	Manuals, Technical: Storage and Maintenance Procedures(for Nonnuclear and Chemical Munitions/Biologi Research Agents)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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